MCO-1D Field Cards — Rev 24

Quick reference for all systems. Keep these with the aircraft.

PRE-FLIGHT

Step 1	Say: "Ready to fly" — AI checks faults, battery, GPS, wing guard.
Step 2	Confirm fuselage width and TPU start if first flight of session.
Step 3	Say: "Arm it" — stand clear — badge turns amber.
Battery	Must be > 21.5V. Below 20V: do not fly.
Wing guard	Must be calibrated. AI will tell you if not.

◆ **NOTE** Never skip the pre-flight check. The AI will stop you if something is wrong.

POWER LEVER

0–15%	Soaring / glide. Variometer activates. Engine effectively off.
15–28%	Low power. Slow flight / approach.
28–45%	Powered cruise.
45–55%	Altitude hold. 50% = exact level.
55–75%	Efficient glide range.
75–87%	Lift dump (braking). Wings shed lift.
87–97%	Shuttle brake. Maximum drag. J-hook at >20m.
97–100%	Emergency deceleration. 2g. Auto-reverts.

REFERENCE BUGS

VSI bug	vsi_ref -100 = 100 ft/min descent. vsi_ref 0 = level.
Heading bug	hdg_ref 090 = East. Shows on DI card.
Altitude bug	alt_ref 1000 = 1000 ft. alt_ref 300 m = 300 metres.
Clear all	vsi_ref_clear / hdg_ref_clear / alt_ref_clear
Status	precision_status — all three bugs + trim quality.

◆ **NOTE** Bugs are display only. The aircraft does not hold any reference value.

FLIGHT MONITOR

Tier 1 — Immediate	Battery <19V: land now. Battery <20V: begin return. Power >4.5kW. Vibration >0.35g. Guard >10/min.
Tier 2 — Advisory	Battery <21.5V. Power >3.5kW. Vibration >0.15g. Trim drift >50 fpm. Rope >2%/min.
Silence rules	15s between alerts. 60s before repeating same alert.
Report	Say: "Monitor report" or "flight health"

◆ **NOTE** Tier 2 suppressed during aerobatic manoeuvres. Tier 1 always fires.

AI OBSERVER

Proposes	~once per flight. Specific, one sentence.
Approve	Say: "Yes" / "go ahead" / "try it"
Decline	Say: "No" / "not now"
Report	Say: "Observer report" / "learning quality"
What it writes	Cascade bounds, genome weights, archive entries — with approval only.

◆ **NOTE** Observer never acts without your yes. Trial runs 4-10 seconds. Reports result.

SOARING / VARIOMETER

Auto-activate	Lever below 15%. Netto variometer lights up.
Basic VSI	Altitude rate. Normal powered flight.
Total energy	Removes speed/altitude trades. Show true energy.
Netto	Air mass only. Zero in still air. Positive = thermal.
Best glide	~9.4 m/s (18 kt). Distance maximised.
Min sink	~7.2 m/s (14 kt). Time in thermal maximised.
MacCready	Shown on GS. Faster between strong thermals.
Thermal	"Soaring status" — strength, heading, MacCready advisory.

◆ **NOTE** Thermal detected when netto > 0.2 m/s for 5+ seconds. Observer announces it.

STABILITY / COORDINATION

Proverse	Roll command yaws nose into turn. No adverse yaw.
Weathercock	Cascade keeps nose tracking through sustained turn.
Dihedral	Wings return toward level after gusts. Hands-off stable.

System learns	All three optimised together via ALSS. Better each session.
Status	Say: "Stability status" / "coordination quality"

DENSITY ALTITUDE

What it does	Estimates air density from pitot + IMU. Scales amplitude.
Band A	ρ/ρ■ 0.75–1.0. Amplitude 96°–108°. Normal ops.
Band B	ρ/ρ■ 0.65–0.75. Amplitude 108°–115°. Mountain ops.
Band C	ρ/ρ■ < 0.65. Amplitude 115° + slight freq increase. Extreme.
Status	Say: "Density altitude status" / "thin air compensation"

◆ **NOTE** Electric motors maintain full power at altitude. Only aerodynamics change.

AFTER FLIGHT

Save	Say: "We're done" — saves all learning and disarms.
Mandatory	Always save before battery disconnect.
Battery	Disconnect only after "Session saved. Safe to disconnect."
Export	Knowledge exports automatically if USB mounted.

◆ **NOTE** Never disconnect battery without saving. All session learning will be lost.